BCS-094	Machine Learning	L T P 3 0 0	3 Credits

Pre-requisites:

Course Objectives:

CO1	understanding of issues and challenges of Machine Learning
CO2	Should be able to select data, model selection, model complexity etc.
CO3	Understanding of the strengths and weaknesses of many popular machine learning approaches.
CO4	To learn about Artificial Neural Networks
CO5	To understand Logistic Regression

Detailed Syllabus

MODULE-I

Introduction to Machine Learning Supervised Learning, Unsupervised Learning, Reinforcement Learning. Probability Basics Linear Algebra Statistical Decision Theory – Regression & Classification Bias – Variance Linear Regression Multivariate Regression.
Dimensionality Reduction Subset Selection, Shrinkage Methods, Principle Components Regression Linear Classification, Logistic Regression, Linear Discriminant Analysis Optimization, Classification-Separating Hyperplanes Classification.

MODULE-II

Artificial Neural Networks (Early models, Back Propagation, Initialization, Training & Validation) Parameter Estimation (Maximum Likelihood Estimation, Bayesian Parameter Estimation) Decision Trees Evaluation Measures, Hypothesis Testing Ensemble Methods, Graphical Models.
Neural Networks, threshold logic units, linear machines, networks of threshold learning units, Training of feed forward networks by back propagations, neural networks vs. knowledge-based systems.

MODULE-III

Clustering, Gaussian Mixture Models, Spectral Clustering Ensemble Methods Learning Theory, Inductive Logic Programming, notation and definitions, introducing recursive programs, inductive logic programming vs decision tree induction.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Identify the characteristics of datasets and compare the trivial data and big data for various applications.
CO2	Understand machine learning techniques and computing environment that are suitable for the applications under consideration.
CO3	Develop scaling up machine learning techniques and associated computing techniques and technologies for various applications
CO4	Implement various ways of selecting suitable model parameters for different machine learning techniques.
CO5	Discriminate inductive logic programming and decision tree induction.

Text Books:

1. T. Hastie, R. Tibshirani, J. Friedman. The Elements of Statistical Learning, 2008.
2. Christopher Bishop. Pattern Recognition and Machine Learning.
3. Andreas, C. Muller & Sarah Guido, O'Reilly Introduction to Machine Learning with Python A guide for data scientists,
4. Nils J.Nilsson, Introduction to Machine learning.